Structural Optimization (B-KUL-H0T88A)

Authors:

Lukas Campaert (r0885501)
Tiebert Lefebure (r0887630)
Mechanical Engineering – Faculty of Engineering Science

Multiresolution Topology Optimization (MTOP)

Computational-efficiency assessment for the half MBB beam

Instructor: Prof. M. Schevenels

Academic Year 2025–2026

Abstract

This report assesses the computational efficiency of the multiresolution topology optimization (MTOP) approach of Nguyen et al. for the two-dimensional minimum-compliance MBB beam. The analysis is performed on a coarse finite element mesh while the design is represented on a finer density mesh. The convergence behavior, wall-clock time and optimized layout are compared with a classical density-based formulation at the same final density resolution. For the assignment specification of 600×200 density cells, the MTOP scheme reproduces the classical layout to within $0.045\,\%$ of fine-mesh compliance for sensitivity filtering and to within $0.04\,\%$ for density filtering, while reducing the optimization time by a factor of 21 for sensitivity filtering and 5 for density filtering. A per-phase wall-clock breakdown confirms that the speedup is generated by the FE solve, which is 40 times faster on the coarse 120×40 analysis mesh than on the classical 600×200 mesh, and is partly absorbed by per-density-cell operations whose cost is identical for both formulations. A separate coarse-coarse baseline with both analysis and density meshes equal to 120×40 reaches a fine-mesh compliance of 226.036, showing that the mechanical benefit of MTOP over a much simpler coarse model is modest for this two-dimensional case, but that MTOP preserves the fine 600×200 design resolution at almost the same cost. The optimality-criteria update is then replaced by MMA and combined with sensitivity filtering, density filtering and Heaviside projection at three threshold values η. Heaviside projection produces near-binary designs whose fine-mesh compliance is about $17\,\%$ lower than the gray density-filtered design at the same volume fraction; the symmetric threshold $\eta = 0.5$ gives the lowest compliance of all the configurations tested. All density-filtered runs are stopped by a compliance-stability criterion rather than by the design-change tolerance, reflecting the well-known incompatibility between the multiplicative OC update and density filtering on a fine mesh.

Contents

1 Problem statement

The objective of the assignment is to assess whether the MTOP approach proposed by Nguyen et al. [2] reduces the computational cost of two-dimensional minimum-compliance topology optimization without altering the optimized layout. Two formulations are compared on the half MBB beam benchmark from the lecture material:

1. a classical density-based formulation in which the analysis mesh and the density mesh coincide, and
2. the MTOP formulation in which the analysis mesh is coarser than the density mesh.

The comparison is carried out for the four assignment steps: optimality criteria with sensitivity filtering (steps 1 and 2), optimality criteria with density filtering (step 3), and the method of moving asymptotes combined with sensitivity filtering, density filtering and Heaviside projection at three threshold values η (step 4). The full experimental setup is given in Section 6.1.

2 Structural model

2.1 Geometry, boundary conditions and loading

The structure is the half MBB beam used in the Chapter 9 lecture code [1]. A unit downward point load is applied to the upper-left corner. The left edge is restrained against horizontal motion (symmetry condition) and the lower-right corner is restrained against vertical motion (roller). Bilinear plane-stress quadrilateral elements with unit edge length, unit thickness, $E_0 = 1$ and Poisson's ratio $\nu = 0.3$ are used for the analysis.

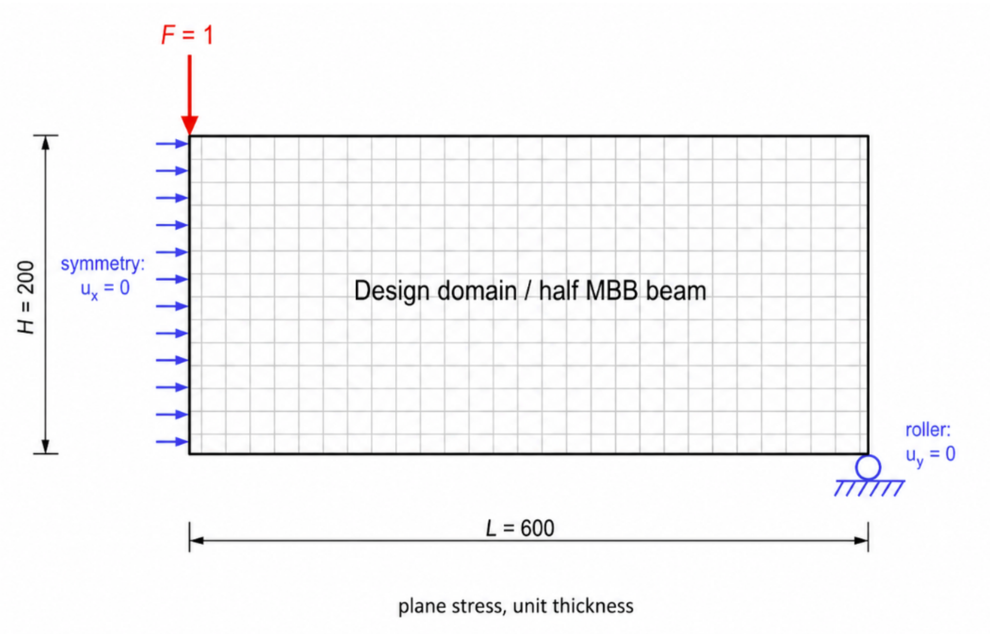


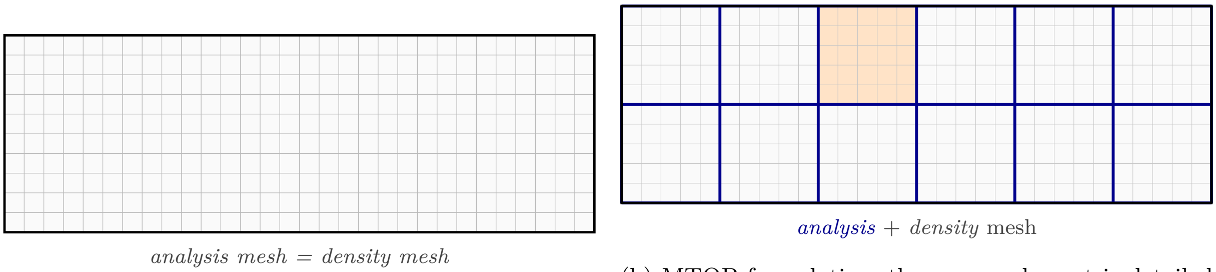
Figure 1: Geometry, boundary conditions and loading of the half MBB beam.

2.2 Discretization

The classical baseline uses a finite element mesh of 600×200 elements. The MTOP model uses a coarse finite element (analysis) mesh of 120×40 elements with 5×5 density cells per analysis element, giving the same 600×200 density resolution as the classical model. The two discretizations are summarized in table 1 and illustrated schematically in fig. 2; the per-element 5×5 density-cell layout that defines the MTOP enrichment is detailed in fig. 3.

Table 1: Discretization parameters of the classical and MTOP formulations.

Model	Analysis (FE) mesh	Density cells per element	Density mesh
Classical	600×200	1×1	600×200
MTOP	120×40	5×5	600×200



(a) Classical formulation.

(b) MTOP formulation; the orange element is detailed in fig. 3.

Figure 2: Classical and MTOP discretizations at aspect ratio 3:1.

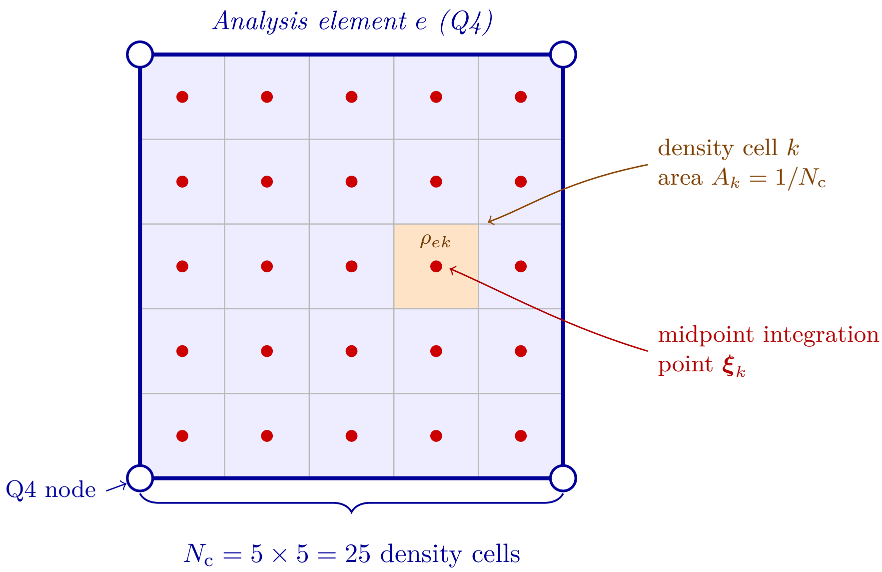


Figure 3: Anatomy of one MTOP analysis element.

The number of free displacement degrees of freedom is $N_{\text{dof}}^{\text{cl}} \approx 2.41 \times 10^5$ for the classical mesh and $N_{\text{dof}}^{\text{mtop}} \approx 9.88 \times 10^3$ for the MTOP analysis mesh. The factor of about 24 in mesh size translates into a much larger reduction in the cost of solving $\mathbf{Ku} = \mathbf{f}$, which dominates the computation in this 2D setting.

2.3 Material interpolation

The Young's modulus depends on the physical density ρ through the SIMP law

$$E(\rho) = E_{\min} + \rho^p \, (E_0 - E_{\min}), \tag{1}$$

with penalization power $p = 3$, $E_0 = 1$ and $E_{\min} = 10^{-9}$. The volume fraction is fixed at $V^* = 0.5$.

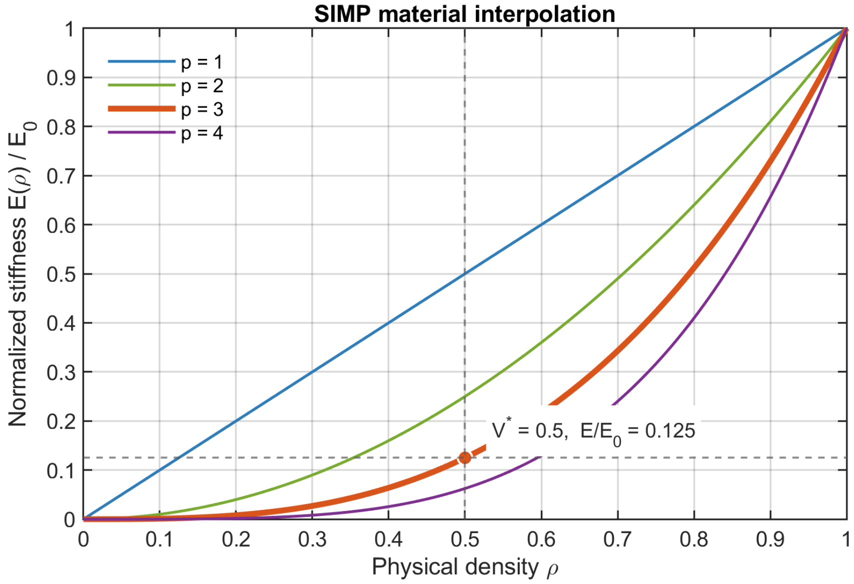


Figure 4: SIMP interpolation between void and solid material for different penalization powers. The report uses $p = 3$, which strongly penalizes intermediate physical densities; at the prescribed volume fraction $\rho = V^* = 0.5$, the normalized stiffness is only $E/E_0 = 0.125$.

3 Optimization problem

Let $\boldsymbol{x} \in [0, 1]^N$ denote the vector of design variables and $\boldsymbol{\rho} = \mathcal{F}(\boldsymbol{x})$ the corresponding physical densities, where $\mathcal{F}$ is the filter operator. The minimum-compliance problem reads

$$\min_{\boldsymbol{x}\in[0,1]^N} \quad C(\boldsymbol{x}) = \mathbf{f}^T\mathbf{u}(\boldsymbol{\rho}), \tag{2}$$

$$\text{s.t.} \quad \mathbf{K}(\boldsymbol{\rho})\,\mathbf{u}(\boldsymbol{\rho}) = \mathbf{f}, \tag{3}$$

$$\frac{1}{N}\sum_{i=1}^{N} \rho_i \leq V^*, \tag{4}$$

where N is the number of density cells. For sensitivity filtering one has $\rho_i = x_i$ and the filter is applied to the compliance sensitivity, while for density filtering $\rho_i = \sum_j H_{ij}x_j / \sum_j H_{ij}$ with cone weights $H_{ij} = \max(0, r_{\min} - \|\mathbf{x}_i - \mathbf{x}_j\|)$.

4 Solution technique

4.1 Classical formulation

The classical baseline is the standard SIMP density-based formulation of Sigmund [3] in the vectorized form of Andreassen et al. [1]. The global stiffness matrix is assembled from the analytical bilinear Q4 element stiffness $\mathbf{K}_0$ as

$$\mathbf{K} = \sum_{e=1}^{N} \big[E_{\min} + \rho_e^p(E_0 - E_{\min})\big]\mathbf{K}_0^{(e)}. \tag{5}$$

The compliance sensitivity follows from the self-adjoint structure of the problem,

$$\frac{\partial C}{\partial \rho_e} = -p\,(E_0 - E_{\min})\,\rho_e^{p-1}\,\mathbf{u}_e^T \mathbf{K}_0 \mathbf{u}_e. \tag{6}$$

Sensitivity filtering with a cone kernel of radius $r_{\min} = 24$ density cells is applied to $\partial C/\partial \rho_e$ following the convention used in the lecture code, see Section 4.3. The volume constraint is enforced by the optimality criteria (OC) update with bisection on the Lagrange multiplier; the move limit is $m = 0.2$ and the convergence tolerance on the maximum design change is $\varepsilon = 0.01$.

4.2 Multiresolution formulation

In the MTOP formulation each analysis element e contains $N_c = 25$ density cells. Following Nguyen et al. [2], the element stiffness matrix is integrated by midpoint quadrature with one integration point at the centre of every density cell:

$$\mathbf{K}_e = \sum_{k=1}^{N_c} \left[E_{\min} + \rho_{ek}^p (E_0 - E_{\min})\right] \mathbf{I}_k, \tag{7}$$

where ρ_{ek} denotes the density of cell k inside element e and

$$\mathbf{I}_k = A_k\, \mathbf{B}^T(\boldsymbol{\xi}_k)\, \mathbf{D}_0\, \mathbf{B}(\boldsymbol{\xi}_k). \tag{8}$$

$\mathbf{B}(\boldsymbol{\xi}_k)$ is the strain–displacement matrix of the bilinear Q4 shape functions evaluated at the cell centre $\boldsymbol{\xi}_k$, $\mathbf{D}_0$ is the plane-stress constitutive matrix at $E_0 = 1$, and $A_k = 1/N_c$ is the area of one density cell. The 25 templates $\mathbf{I}_k$ are precomputed once per run; their sum approximates the analytical Q4 stiffness $\mathbf{K}_0$ to the order of the midpoint rule.

The compliance sensitivity with respect to the density of cell k in element e follows directly from the chain rule applied to eq. (7),

$$\frac{\partial C}{\partial \rho_{ek}} = -p\,(E_0 - E_{\min})\,\rho_{ek}^{p-1}\,\mathbf{u}_e^T\, \mathbf{I}_k\, \mathbf{u}_e. \tag{9}$$

This expression replaces the element-wise strain-energy density of eq. (6) by a per-cell quantity that depends on the local position of the cell within the analysis element. The volume sensitivity is uniform: $\partial V/\partial \rho_{ek} = 1/N$.

4.3 Filtering

The cone kernel $H_{ij} = \max(0, r_{\min} - \|\mathbf{x}_i - \mathbf{x}_j\|)$ acts on the density grid and is identical for the classical and MTOP runs. With the lecture convention used in `ex1a.m`, the sensitivity filter applied to eq. (6) or eq. (9) reads

$$\widetilde{\frac{\partial C}{\partial x_i}} = \frac{1}{\max(\gamma, x_i)} \frac{\sum_j H_{ij}\, x_j\, (\partial C/\partial x_j)}{\sum_j H_{ij}}, \qquad \gamma = 10^{-3}. \tag{10}$$

A filter radius of $r_{\min} = 24$ density cells is used both for the classical mesh and for the MTOP density mesh, so that the two formulations enforce the same minimum length scale.

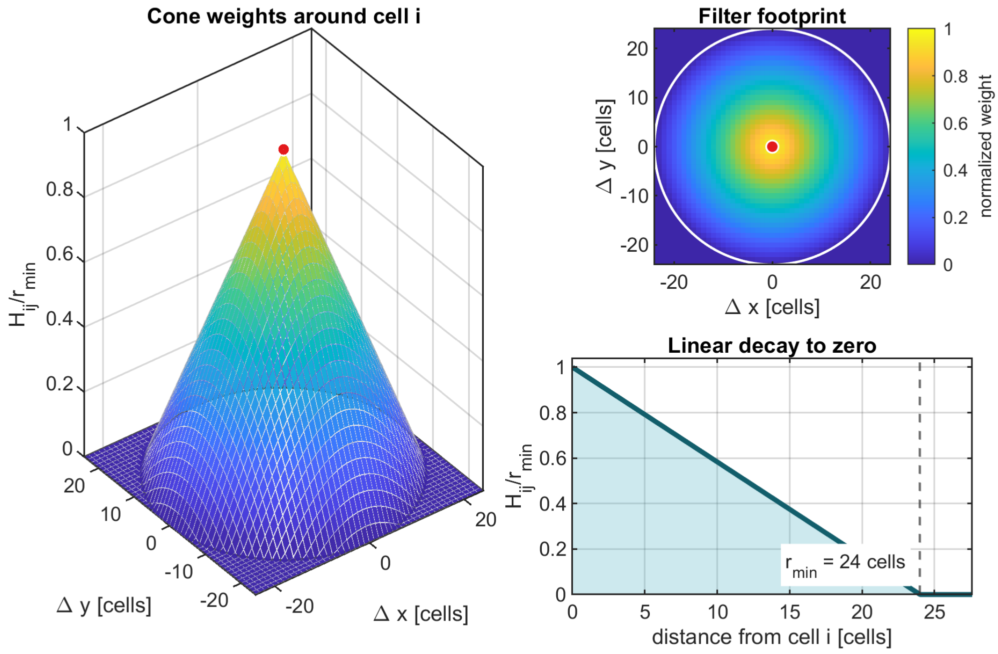


Figure 5: Cone-kernel filter used in all experiments. The central cell receives weight $r_{\min}$, neighboring cells receive weights that decay linearly with distance, and cells at or beyond $r_{\min} = 24$ density cells receive zero weight. The same stencil is applied on the 600×200 density grid for both the classical and MTOP formulations.

4.4 Optimality criteria update

The OC update is identical for both formulations and uses the standard form

$$x_i^{\text{new}} = \max\big[0,\, \max(x_i - m,\, \min(1,\, \min(x_i + m,\, x_i\sqrt{B_i/\lambda})))\big], \qquad B_i = -\frac{\widetilde{\partial C/\partial x_i}}{\partial V/\partial x_i}, \tag{11}$$

with $m = 0.2$ and λ adjusted by bisection until the volume constraint is active.

4.5 Convergence criteria

All runs use the standard design-change criterion $\max_i |\Delta x_i| < \varepsilon$ with $\varepsilon = 0.01$, but the density-filtered runs (Section 7.3, 7.4, 7.5) also accept a compliance-plateau criterion: the optimization terminates if the relative spread of the compliance over a sliding window of 20 iterations is below $\varepsilon_C = 10^{-4}$. This second criterion captures the well-known incompatibility between the multiplicative OC update and density filtering on a fine mesh (Section 8.5): the filter spreads each design-variable change over a neighbourhood of cells, but the OC update rescales every variable individually with no built-in damping, so high-frequency components of the design alternate sign from one iteration to the next even though the compliance has long since plateaued. With only the design-change criterion these runs hit their iteration cap; the compliance-stability criterion lets them stop cleanly. The maximum iteration counts are 500 for the OC and MMA runs and 800 for the Heaviside continuation runs.

4.6 MMA and Heaviside projection

In step 4 the OC update is replaced by the method of moving asymptotes (MMA) of Svanberg [4], which approximates the objective and constraint by separable convex functions of moving-asymptote-shifted variables and solves the resulting subproblem by an interior-point method. The implementation `mma.m` that ships with the lecture toolbox is used unmodified. The objective passed to MMA is $f_0 = C/100$ and the volume constraint is $f_1 = v/V^* - 1 \le 0$; the corresponding gradients are $\partial C/\partial x/100$ and $\partial v/\partial x/V^*$. The design variables remain in $[0,1]$ as before. The geometric idea is illustrated in fig. 6: at each iterate x_k the algorithm builds a separable convex approximation $\tilde{f}_k(x)$ that matches f and ∇f at x_k and diverges to $+\infty$ at the lower and upper asymptotes L_k and U_k; the next iterate x_{k+1} is the minimizer of $\tilde{f}_k$ over the trust region (L_k, U_k). The asymptotes are then re-centered around x_{k+1}, contracting when the iterate oscillates and expanding when it progresses monotonically, which gives MMA its adaptive trust-region behavior.

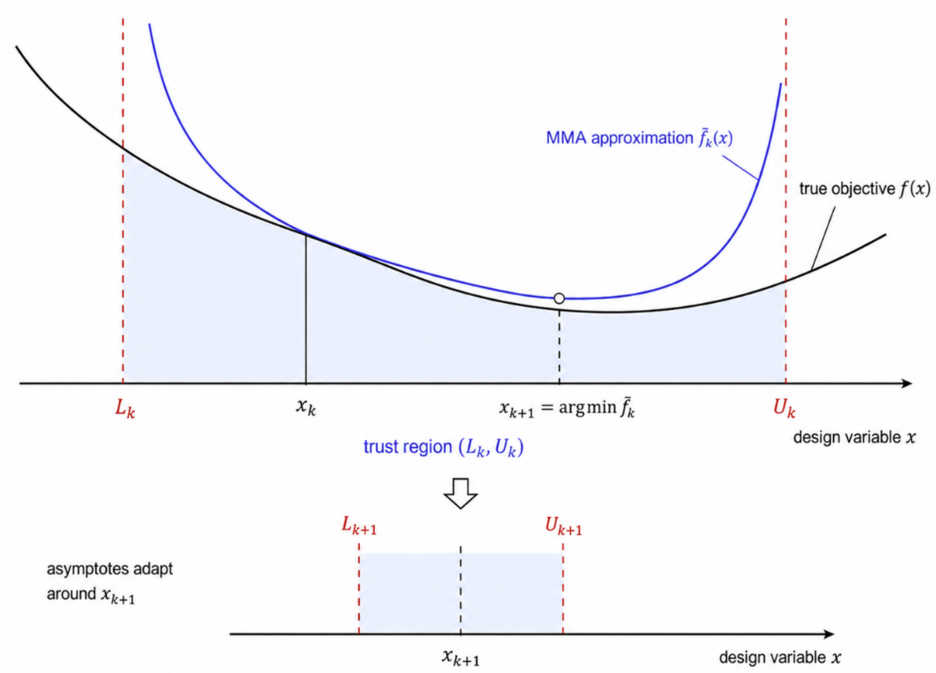


Figure 6: One MMA outer iteration. Top: the separable convex approximation $\tilde{f}_k$ (blue) and its minimizer x_{k+1} on the trust region (L_k, U_k). Bottom: asymptotes adapted around x_{k+1} for the next subproblem.

For the Heaviside-projection variant, density filtering is followed by a smooth projection of Wang, Lazarov and Sigmund [5] towards $\{0, 1\}$,

$$\rho_i = \frac{\tanh(\beta\eta) + \tanh\big(\beta(\tilde{\rho}_i - \eta)\big)}{\tanh(\beta\eta) + \tanh\big(\beta(1-\eta)\big)}, \tag{12}$$

where $\tilde{\rho}_i = \mathcal{F}(\boldsymbol{x})_i$ is the density-filtered field, $\eta \in (0,1)$ is the projection threshold and $\beta \ge 0$ controls the projection sharpness. The compliance and volume sensitivities pick up the additional factor

$$\frac{\partial \rho_i}{\partial \tilde{\rho}_i} = \frac{\beta\,\mathrm{sech}^2\big(\beta(\tilde{\rho}_i - \eta)\big)}{\tanh(\beta\eta) + \tanh(\beta(1-\eta))}, \tag{13}$$

which multiplies $\partial C/\partial \rho_i$ before the linear filter chain rule is applied to obtain $\partial C/\partial x_k$. Continuation on β is used to avoid local minima at high projection sharpness: β starts at 1 and is doubled every 50 iterations until reaching $\beta_{\max} = 16$.

5 Sensitivity verification

The analytical sensitivities are verified with three finite-difference strategies (component-wise, directional and Taylor remainder) applied to three derivative chains. The first chain verifies the classical SIMP/FE sensitivity $\partial C/\partial \rho_e$ with the filter bypassed. The second verifies the MTOP per-cell sensitivity $\partial C/\partial \rho_{ek}$ of (7)–(9). The third verifies the full step 4 chain: the density filter $\widetilde{\rho} = \mathcal{F}(\boldsymbol{x})$, the Heaviside projection $\rho = \mathrm{H}(\widetilde{\rho};\beta,\eta)$ with representative $\beta = 8$ and $\eta = 0.5$, the MTOP equilibrium analysis, and the chain rule of eq. (13). All checks are run on a reduced MBB instance (12×4 analysis elements, 5×5 density cells per element) with random densities perturbed about the volume fraction, using the same compliance and volume routines as the optimization loop. The volume sensitivity is recovered to machine precision for the classical and MTOP per-cell checks.

Table 2: Component-wise finite-difference verification for two representative design variables per chain (from a sample of six), using forward differences at $h = 10^{-6}$. The full step-size sweep is shown in fig. 7.

Check	Variable	Analytical	Finite difference	Relative error
Classical SIMP	x_6	-86.085230	-86.084777	5.26×10^{-6}
Classical SIMP	x_{26}	-29.992941	-29.992734	6.91×10^{-6}
MTOP per-cell	x_{605}	-2.829491	-2.829567	2.69×10^{-5}
MTOP per-cell	x_{588}	-0.830622	-0.830587	4.22×10^{-5}
Full chain	x_{605}	-15.821941	-15.821666	1.74×10^{-5}
Full chain	x_{588}	-6.819617	-6.819446	2.50×10^{-5}

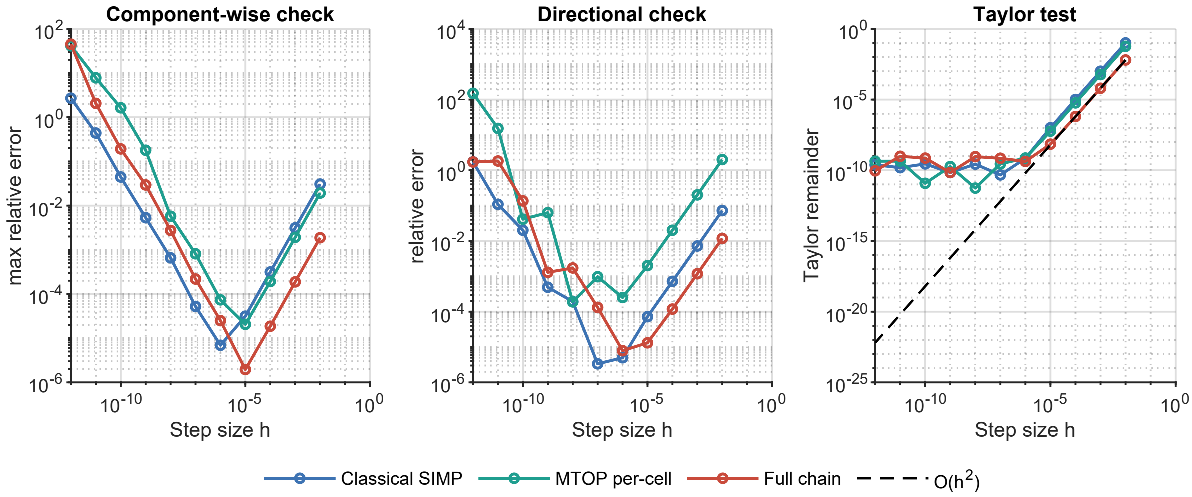


Figure 7: Finite-difference verification across step sizes $h = 10^{-2}, \ldots, 10^{-12}$ for the three derivative chains. Each panel reports the component-wise (max over six samples), directional and Taylor remainder checks; the Taylor remainder follows the expected $\mathcal{O}(h^2)$ slope in the intermediate range.

Table 3: Comparison of the three sensitivity-verification strategies. The component-wise and directional columns report the smallest relative error attained over the step-size sweep; the Taylor slope is fitted over the intermediate range $10^{-3} \geq h \geq 10^{-5}$, where neither truncation nor round-off errors dominate.

Check	Best component-wise error	Best directional error	Taylor slope
Classical SIMP	6.91×10^{-6}	3.35×10^{-6}	2.000
MTOP per-cell	2.06×10^{-5}	1.90×10^{-4}	2.000
Full chain	1.96×10^{-6}	7.88×10^{-6}	1.999

The Taylor remainder test is the strongest acceptance criterion because it verifies both that the directional derivative converges to the finite-difference value and that the first-order Taylor remainder decays quadratically; in all three chains the fitted slope is essentially 2.000. The component-wise check is retained because it localizes any indexing or chain-rule error to a specific design variable. The slightly larger component-wise errors of the MTOP per-cell check reflect the accumulation of $N_c = 25$ midpoint contributions per analysis element, but remain well below any threshold that could affect the optimization trajectory.

6 Numerical experiments

6.1 Experimental setup

Table 4: Numerical experiments performed in this report. Cases 1–4 correspond to assignment steps 1–3 (steps 1 and 2 use sensitivity filtering, step 3 uses density filtering for both the classical and the MTOP runs); cases 5–7 correspond to step 4 with three filter or projection choices, and case 7 is run for three values of the Heaviside threshold η.

Case	Approach	Optimizer	Filter / projection
1	Classical	OC	Sensitivity filter
2	MTOP	OC	Sensitivity filter
3	Classical	OC	Density filter
4	MTOP	OC	Density filter
5	MTOP	MMA	Sensitivity filter
6	MTOP	MMA	Density filter
7	MTOP	MMA	Heaviside projection, $\eta \in \{0.3, 0.5, 0.7\}$

6.2 Quantities recorded per iteration

For each run the iteration counter, the elapsed wall-clock time, the compliance, the volume fraction and the maximum design change are stored. The wall-clock timings reported below are the values stored in the result files generated on one machine and MATLAB release. Absolute timings can change on a rerun, so the speedup factors are interpreted as run-specific comparisons rather than hardware-independent constants.

7 Results

7.1 Classical OC baseline

The classical OC baseline of step 1 converges in 94 iterations. The final compliance is $C_{\mathrm{cl}} = 224.596$, the volume fraction is 0.500 and the maximum design change at convergence is 9.95×10^{-3}. The wall-clock time is 263.23 s, of which 82 % is spent inside the sparse-Cholesky solve of $\mathbf{Ku} = \mathbf{f}$ on the 600×200 analysis mesh. The optimized layout (fig. 8) shows the typical truss-like topology of the half MBB beam: a top chord in compression near the load line, a bottom chord in tension near the support line, and a system of inclined web members carrying the shear toward the support. The convergence histories as functions of the iteration number and of the wall-clock time are shown in figs. 9 and 10.

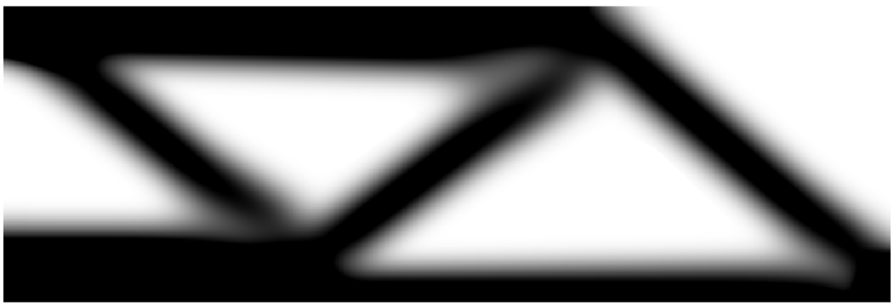


Figure 8: Optimized classical OC design with sensitivity filtering on a 600×200 mesh, $V^* = 0.5$, $p = 3$ and $r_{\min} = 24$ density cells. Black represents solid material, white represents void.

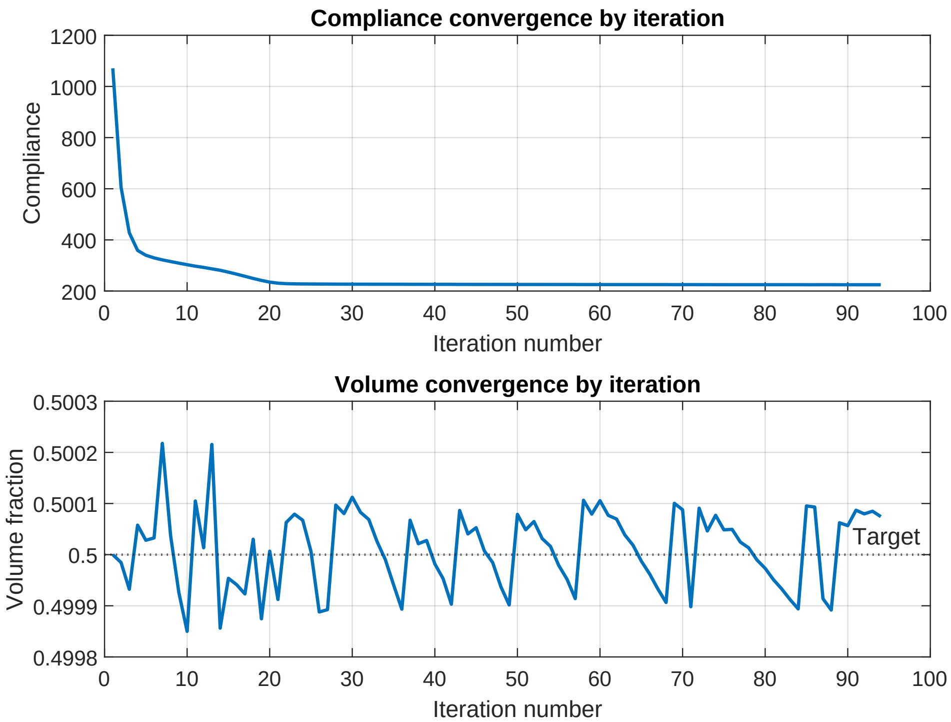


Figure 9: Classical OC convergence history versus iteration number: compliance (top) and volume fraction (bottom).

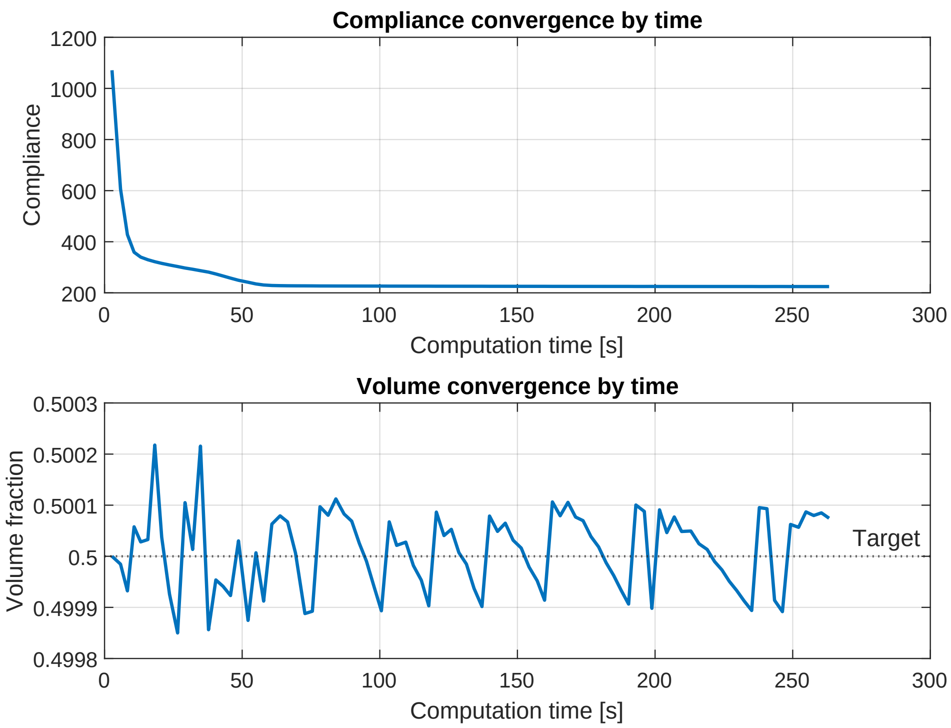


Figure 10: Classical OC convergence history versus elapsed wall-clock time: compliance (top) and volume fraction (bottom).

7.2 MTOP OC with sensitivity filtering

The MTOP OC run reaches the same convergence tolerance in 95 iterations, essentially the same iteration count as the classical baseline, and preserves the truss-like load path of the classical design (fig. 12). The MTOP compliance evaluated on the coarse 120×40 analysis mesh with the multiresolution stiffness matrix (7) is $C_{\mathrm{mtop}}^{\mathrm{coarse}} = 219.069$. To ensure a comparable measure of mechanical performance, the final MTOP density field is also assembled and analyzed on the same 600×200 classical finite element model: this gives a fine-mesh compliance of $C_{\mathrm{mtop}}^{\mathrm{fine}} = 224.696$, only $0.045\,\%$ above the classical compliance C_{cl}. The volume fraction at convergence is 0.500 and the wall-clock time is 12.28 s, a speedup of 21.4 relative to the classical baseline.

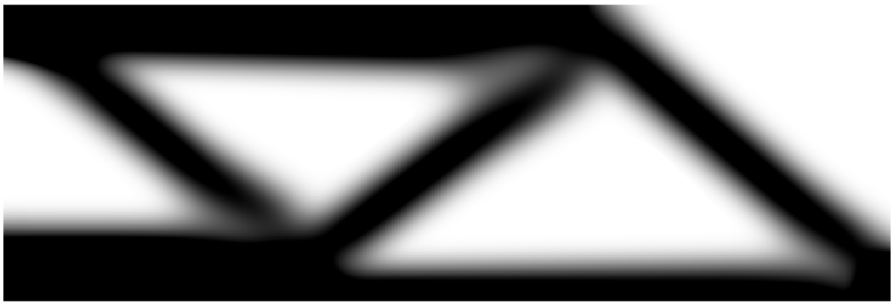


Figure 11: Optimized MTOP OC design with a 120×40 analysis mesh and 5×5 density cells per analysis element.

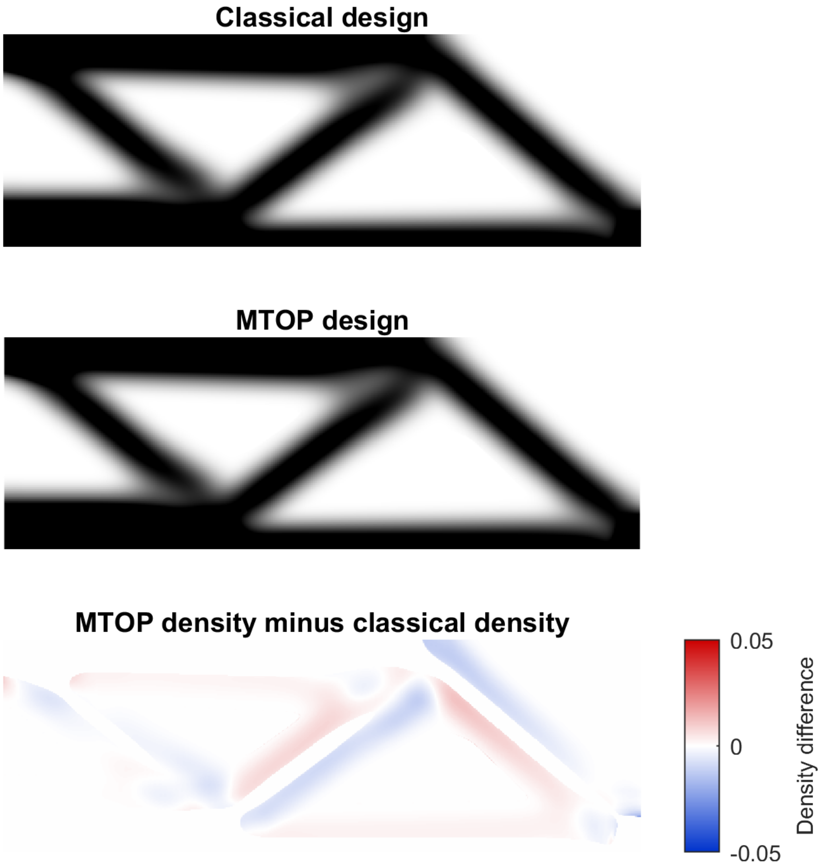


Figure 12: Classical and MTOP OC designs (top, middle) and signed density difference $\rho_{\mathrm{mtop}} - \rho_{\mathrm{cl}}$ (bottom). Maximum, RMS and mean absolute differences: 0.0222, 0.0027, 0.0013.

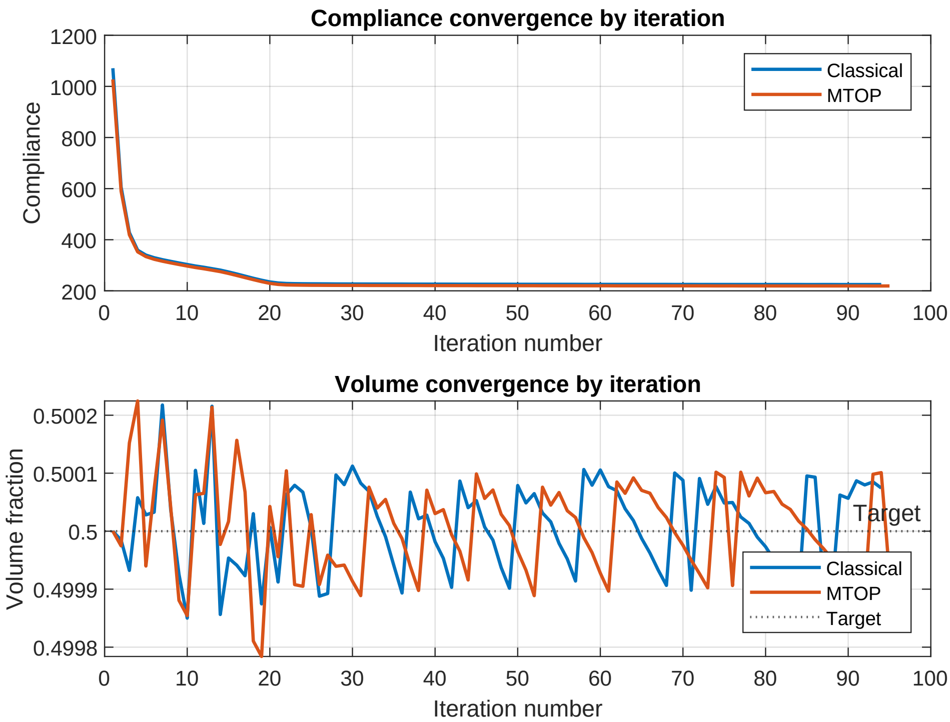


Figure 13: Classical and MTOP OC convergence histories versus iteration number. The two histories are nearly indistinguishable.

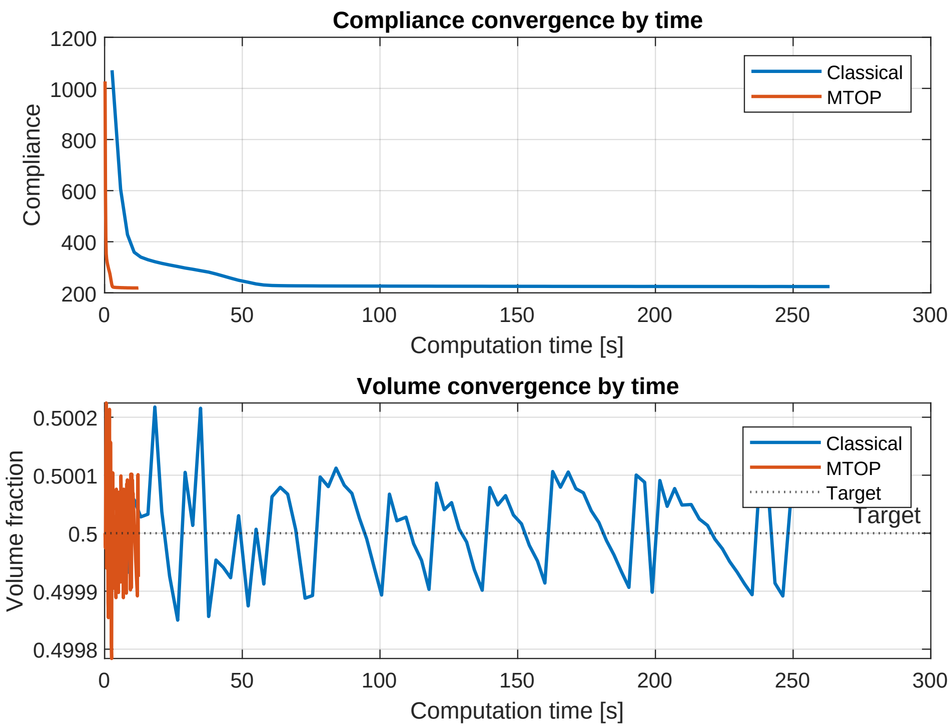


Figure 14: Classical and MTOP OC convergence histories versus wall-clock time; the MTOP curve is compressed against the left axis.

7.3 Classical OC with density filtering

The third experiment replaces the sensitivity filter by a density filter while keeping all other parameters of step 1 unchanged. The compliance drops to a stable plateau within roughly fifty iterations, while the maximum design change stagnates between 0.04 and 0.07 and the optimizer cycles between two near-equivalent layouts; this is the well-known incompatibility of the multiplicative OC update with density filtering, discussed in Section 8.5. The compliance-stability criterion of Section 4.5 therefore terminates the run after 237 iterations; the design-change criterion is not satisfied. The final compliance is $C_{\mathrm{cl}}^{\mathrm{dens}} = 241.083$, the volume fraction is 0.500 and the wall-clock time is 692.91 s. The corresponding design (fig. 15) shows the same load-path topology as the sensitivity-filtered design but with notably wider grey transitions across each member boundary, which is the hallmark of density filtering: the physical density is a smoothed version of the design variable, so the boundary cannot be sharper than one filter radius.

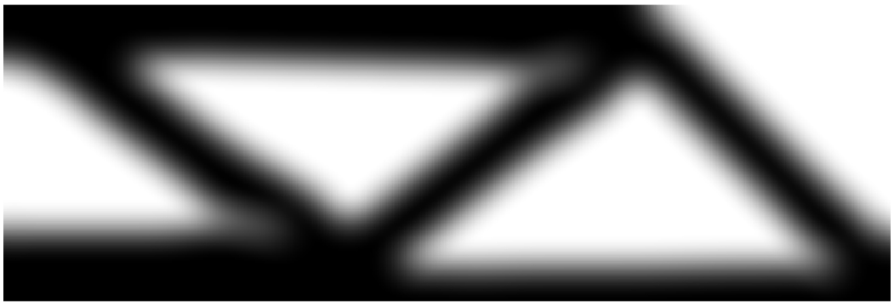


Figure 15: Optimized classical OC design with density filtering on a 600×200 mesh, $V^* = 0.5$, $p = 3$ and $r_{\min} = 24$ density cells.

7.4 MTOP OC with density filtering

The MTOP density run terminates the same way after 299 iterations: the maximum design change does not reach $\varepsilon = 0.01$, but the compliance plateau is detected. The native (multiresolution) compliance at termination is $C_{\mathrm{mtop}}^{\mathrm{coarse,dens}} = 235.363$ and the corresponding fine-mesh compliance is $C_{\mathrm{mtop}}^{\mathrm{fine,dens}} = 240.984$, 0.041 % below the classical density-filtered compliance. The volume fraction at termination is 0.500 and the wall-clock time is 136.57 s, a speedup of 5.07 relative to the classical density-filtered run. The compliance sensitivity fields at the final density-filtered designs (fig. 20) confirm that the classical and MTOP analyses recover the same load-carrying structure; the residual design-variable change is only a high-frequency boundary oscillation, which is why the compliance-plateau criterion provides a cleaner stop than the design-change tolerance.

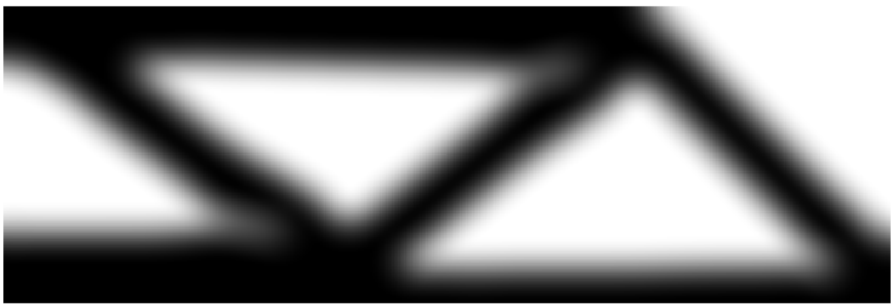


Figure 16: Optimized MTOP OC design with density filtering, on a 120×40 analysis mesh and 5×5 density cells per analysis element.

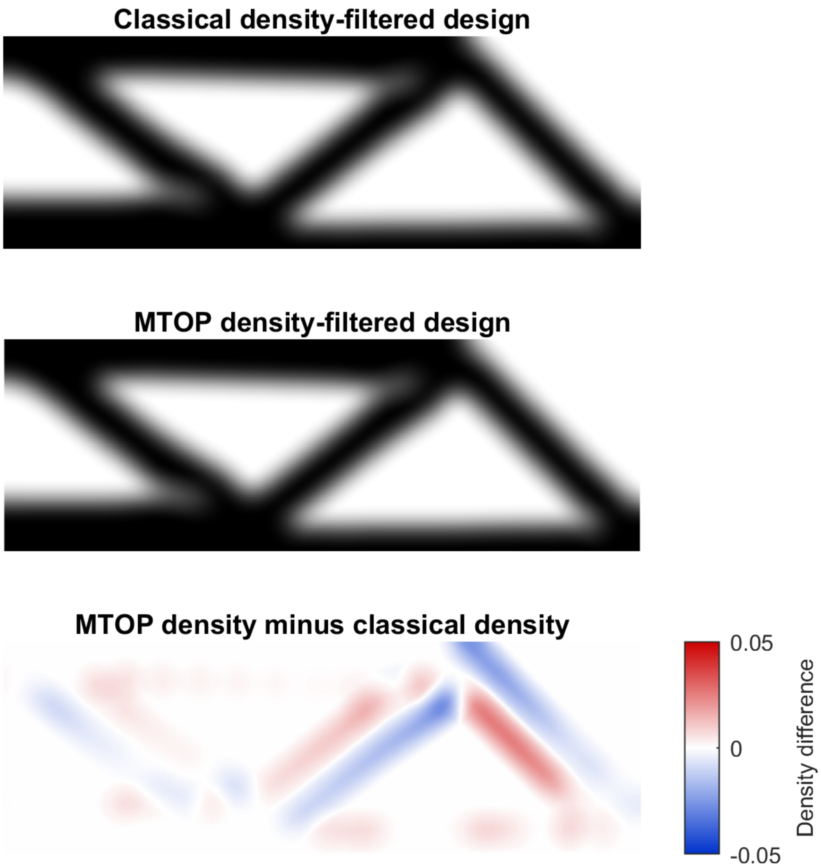


Figure 17: Classical and MTOP OC designs with density filtering (top, middle) and signed density difference (bottom). Maximum, RMS and mean absolute differences: 0.0065, 0.0010, 0.0005, markedly smaller than under sensitivity filtering (fig. 12).

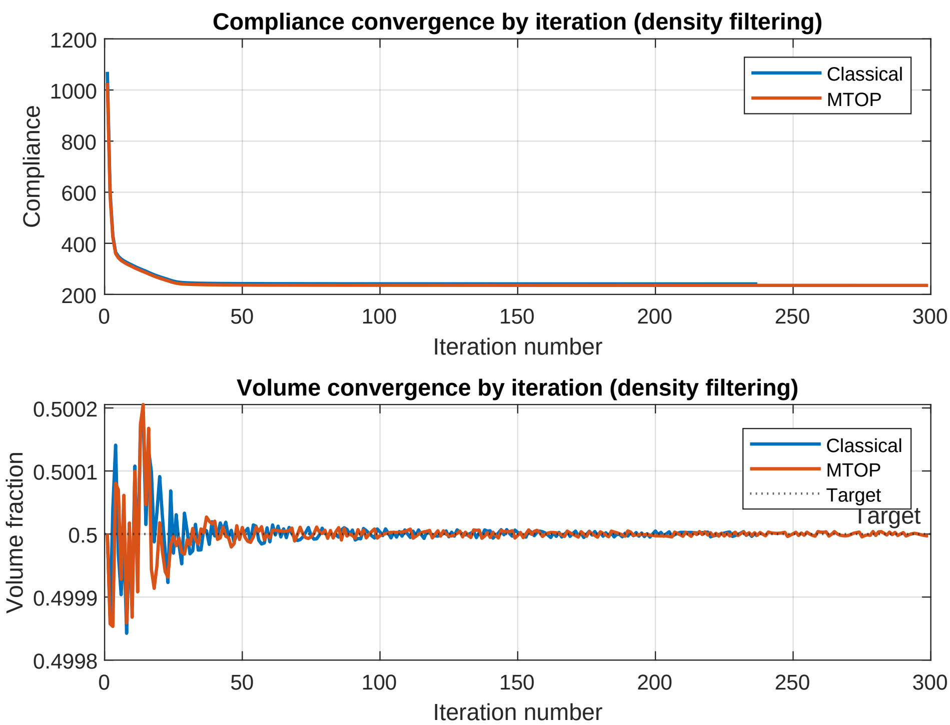


Figure 18: Classical and MTOP OC convergence histories versus iteration number, for density filtering.

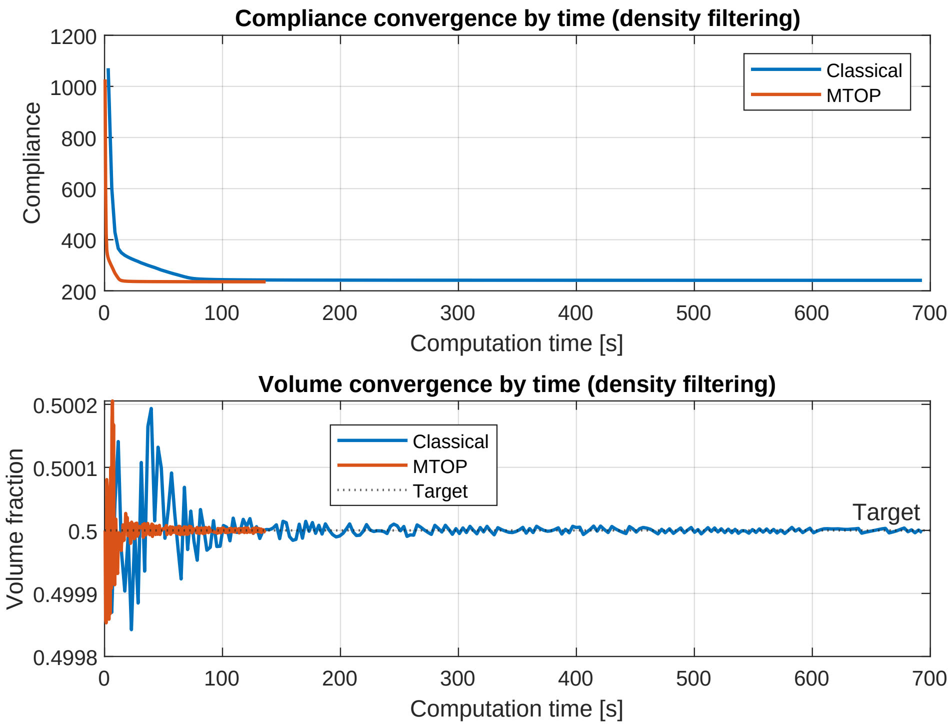


Figure 19: Classical and MTOP OC convergence histories versus wall-clock time, for density filtering.

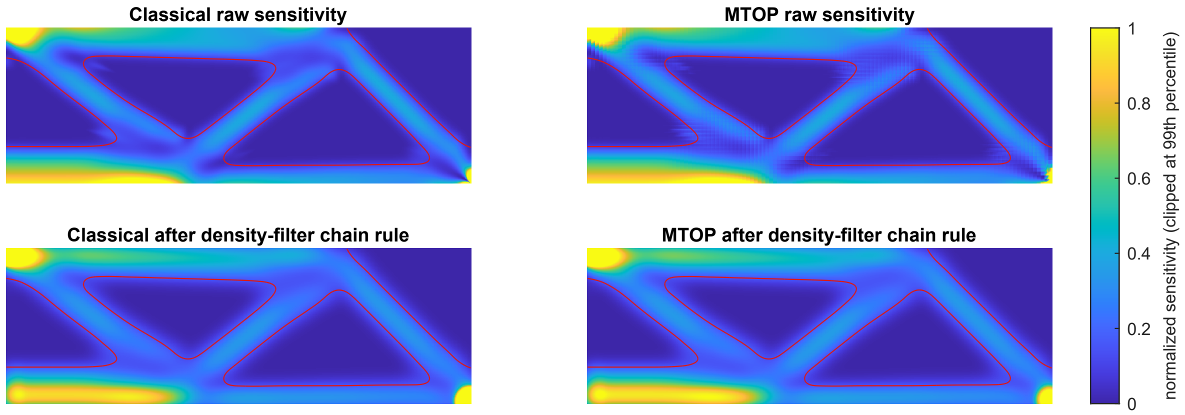


Figure 20: Compliance sensitivity fields at the final density-filtered OC designs, normalized per panel and clipped at the 99th percentile. The red contour marks the $\rho = 0.5$ design boundary.

7.5 MTOP MMA: sensitivity and density filtering

Step 4 keeps the multiresolution analysis mesh of step 2 and the same filter radius, but replaces the OC update by MMA. The first two MMA experiments use sensitivity filtering and density filtering, which are directly comparable with the corresponding OC runs of steps 2 and 3.

With sensitivity filtering, MMA satisfies the change tolerance $\varepsilon = 0.01$ in 250 iterations and gives a final compliance of $C^{\text{coarse}}_{\text{mma,sens}} = 220.875$ on the coarse analysis mesh and $C^{\text{fine}}_{\text{mma,sens}} = 226.502$ on the fine 600×200 analysis mesh. The volume fraction at convergence is 0.500 and the wall-clock time is 187.28 s. The MTOP MMA design is therefore very close to its OC counterpart of step 2, but reached after roughly two and a half times as many iterations and with a slightly higher fine-mesh compliance ($+0.80\,\%$ vs. $C^{\text{fine}}_{\text{mtop}} = 224.696$).

With density filtering, MMA reaches the same compliance plateau as the OC density-filtered runs and is again stopped by the stability criterion after 419 iterations. The compliance is $C^{\text{coarse}}_{\text{mma,dens}} = 234.455$ on the coarse analysis mesh and $C^{\text{fine}}_{\text{mma,dens}} = 240.077$ on the fine analysis mesh. The volume fraction at termination is 0.500 and the wall-clock time is 253.89 s. The fine-mesh compliance lies 0.42 % below the OC density-filtered value from step 3, so MMA produces a marginally better design, but the change-tolerance issue of OC density is not resolved by switching the optimizer alone.

Figure 21: Optimized MTOP MMA design with sensitivity filtering.

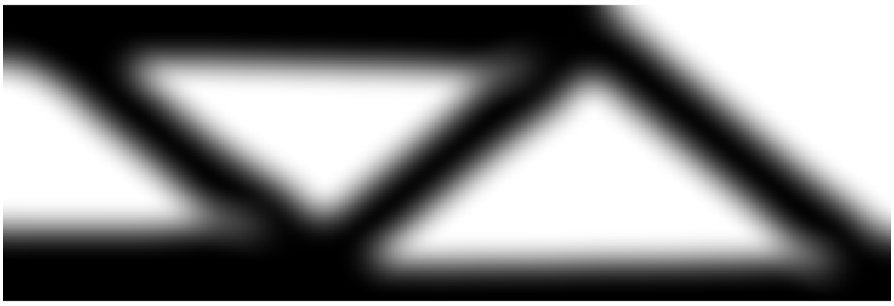


Figure 22: Optimized MTOP MMA design with density filtering.

7.6 MTOP MMA with Heaviside projection

Heaviside projection is applied on top of density filtering with three threshold values $\eta \in \{0.3, 0.5, 0.7\}$. For each η, the continuation strategy described in Section 4.6 is used, doubling β from 1 to 16 every 50 iterations. The three runs reach fine-mesh compliances of 200.016, 199.326 and 200.633 for $\eta = 0.3$, 0.5 and 0.7 respectively, in 419, 563 and 522 iterations; the optimized designs are shown in fig. 23 and all results enter the runtime summary in fig. 27.

$\eta = 0.3$ (solid-biased threshold) $\eta = 0.5$ (symmetric threshold) $\eta = 0.7$ (void-biased threshold)

Figure 23: Optimized MTOP MMA designs with Heaviside projection for the three threshold values η used in step 4.

The corresponding histories for the asymmetric thresholds $\eta = 0.3$ and $\eta = 0.7$ (figs. 25 and 26) follow the same continuation pattern: each β doubling produces a small upward step in compliance followed by a re-optimization plateau. The post-$\beta_{\max}$ tail is shorter for $\eta = 0.3$ (the projection is biased towards solid, which the volume constraint already favours when $V^* = 0.5$) and longer for $\eta = 0.7$ (the projection is biased towards void, so the optimizer has to compensate for longer before the design change drops below the tolerance).

7.7 Runtime comparison

The runtime summary in fig. 27 groups the nine runs by filtering or projection family: sensitivity filtering, density filtering and Heaviside projection. Sensitivity- and density-filtered runs solve

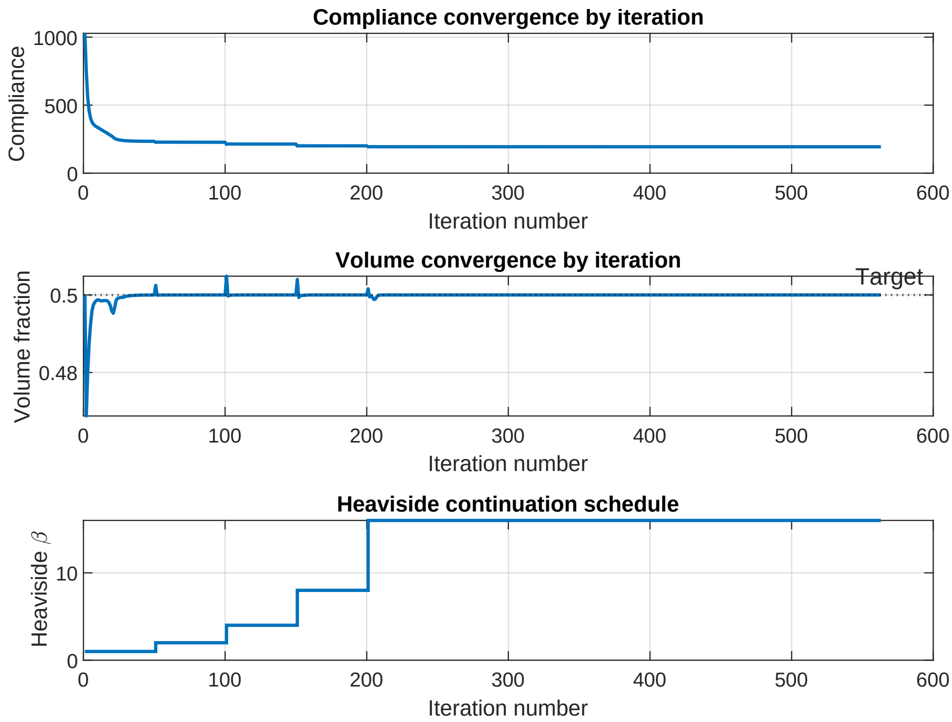


Figure 24: Convergence history for the symmetric Heaviside run ($\eta = 0.5$): compliance, volume fraction and the continuation parameter β versus iteration number.

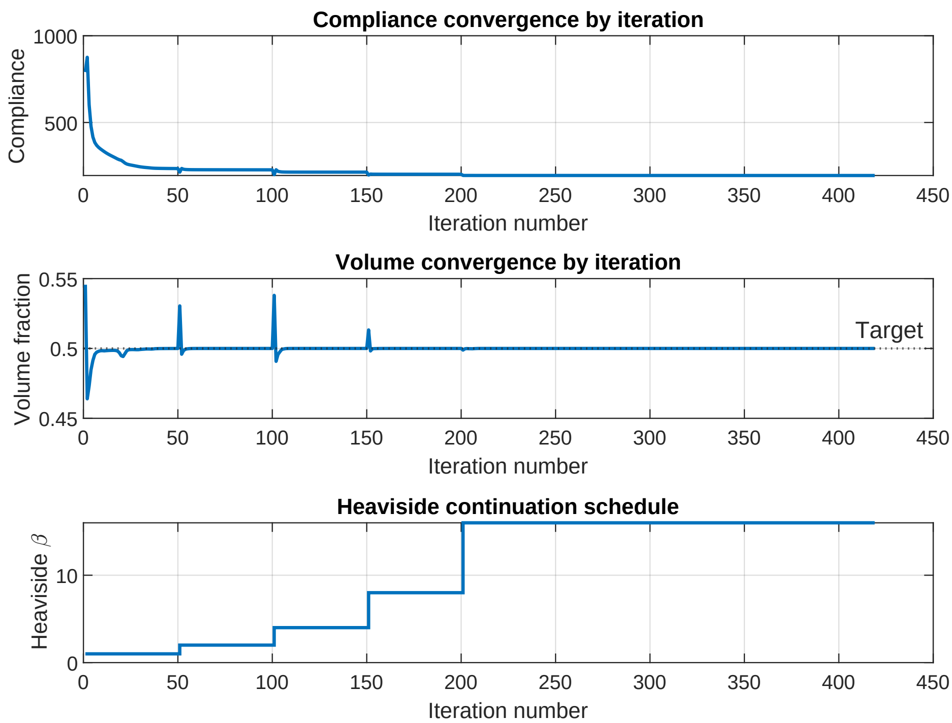


Figure 25: Convergence history for the solid-biased Heaviside run ($\eta = 0.3$): compliance, volume fraction and the continuation parameter β versus iteration number.

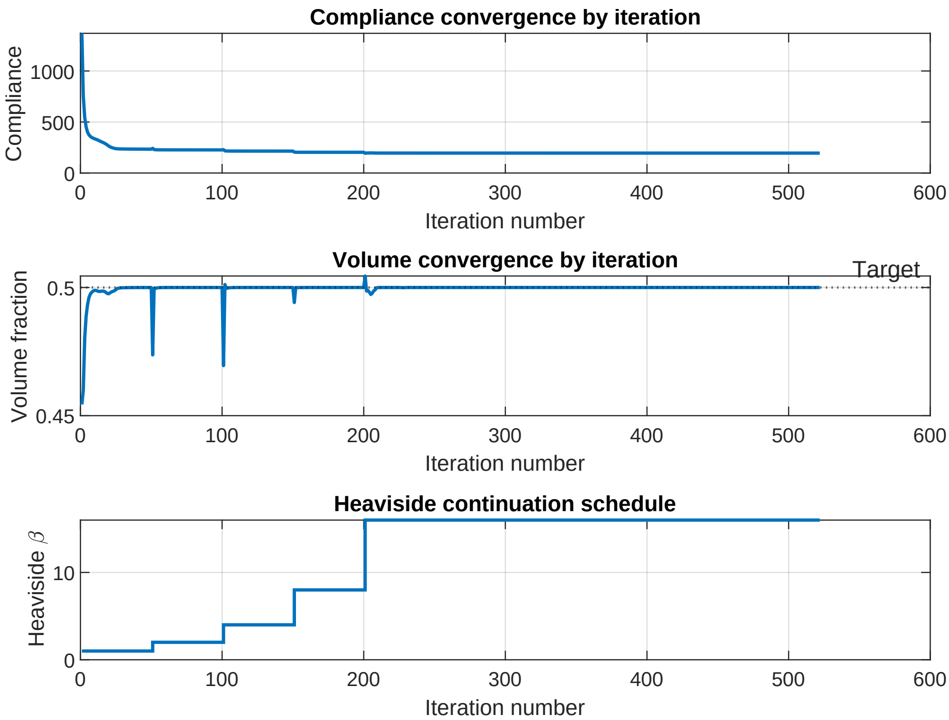


Figure 26: Convergence history for the void-biased Heaviside run ($\eta = 0.7$): compliance, volume fraction and the continuation parameter β versus iteration number.

different optimization problems, so cost comparisons within each family are meaningful but cross-family compliance differences should not be read as algorithmic improvements. The fine-mesh compliance is the comparison value throughout; the native MTOP compliance is reported only as a coarse-analysis diagnostic. Within each family MTOP delivers a $5.07\times$ to $21.4\times$ wall-clock speedup over the classical analysis at essentially identical mechanical performance, and Heaviside projection reduces the fine-mesh compliance by roughly 17 % relative to the density-filtered MMA run at a comparable wall-clock budget.

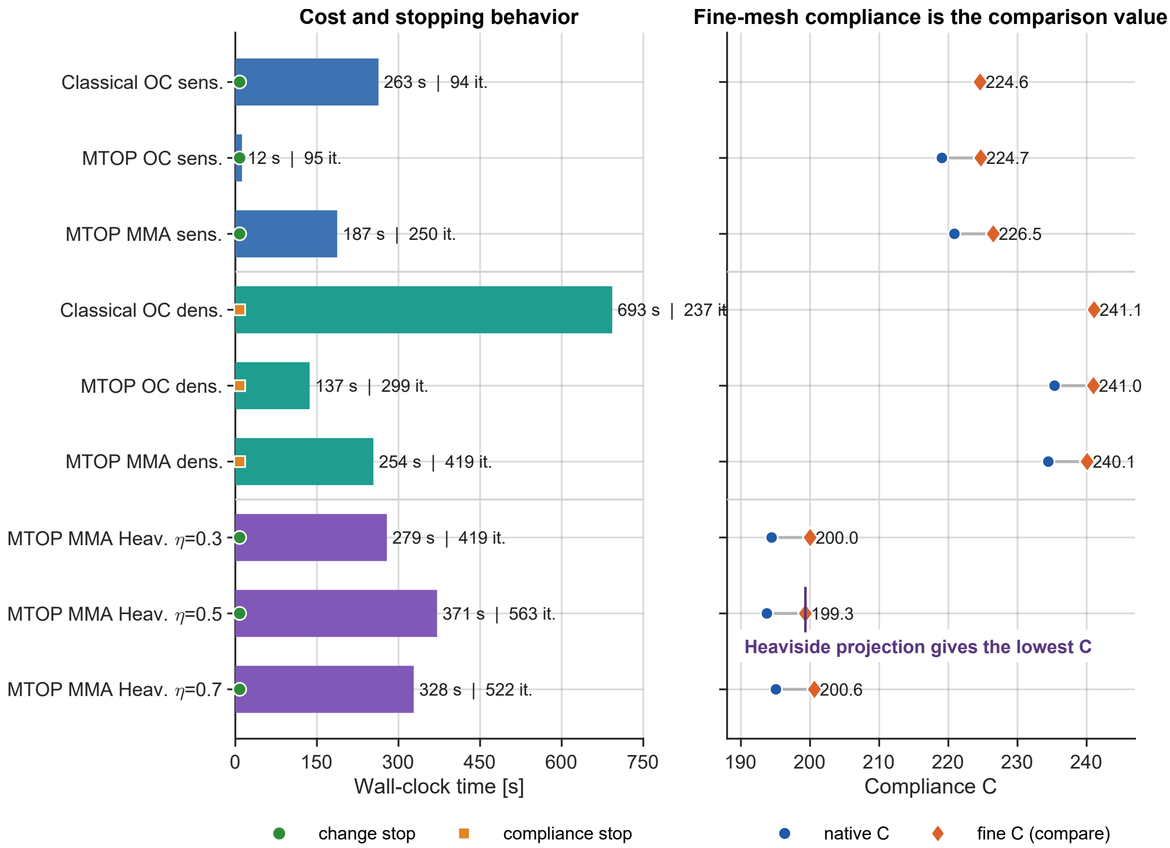


Figure 27: Runtime and compliance summary for the nine runs, sorted by filtering or projection family. The left panel shows cost, iterations and stopping criterion; the right panel reports fine-mesh compliance with the native MTOP compliance as a secondary diagnostic.

8 Discussion

8.1 Convergence behavior

The classical and MTOP OC runs with sensitivity filtering follow almost the same iteration history and converge in essentially the same number of iterations (94 and 95). The volume constraint is also enforced to the prescribed value in both cases. The main effect of MTOP is therefore not a different optimization trajectory, but a lower cost per iteration through the smaller equilibrium problem.

8.2 Topological similarity of the two designs

The MTOP and classical layouts have the same load path: a top chord, a bottom chord and a triangulated web (figs. 8 and 11). The difference plot in fig. 12 shows only small boundary-level redistributions, and the fine-mesh compliance of the MTOP design exceeds the classical one by only 0.0447 %. This confirms that, for sensitivity filtering with $r_{\min} = 24$ density cells, the coarse 120×40 analysis mesh still resolves the dominant structural response.

8.3 Origin of the speedup

The wall-clock advantage of MTOP comes mainly from replacing the 600×200 equilibrium solve by a 120×40 solve. The total speedup is smaller than the reduction in analysis problem size would suggest, because several parts of each MTOP iteration still scale with the fine density resolution: the cone-kernel filter, the assembly of the multiresolution stiffness with 25 density-cell contributions per analysis element, and the per-density-cell strain-energy evaluation (9).

To make this concrete, the cumulative wall-clock time spent in each phase of the iteration was measured separately for FE assembly, FE solve, filtering or chain-rule work, and the optimizer update. fig. 28 visualizes the resulting breakdown for seven representative cases.

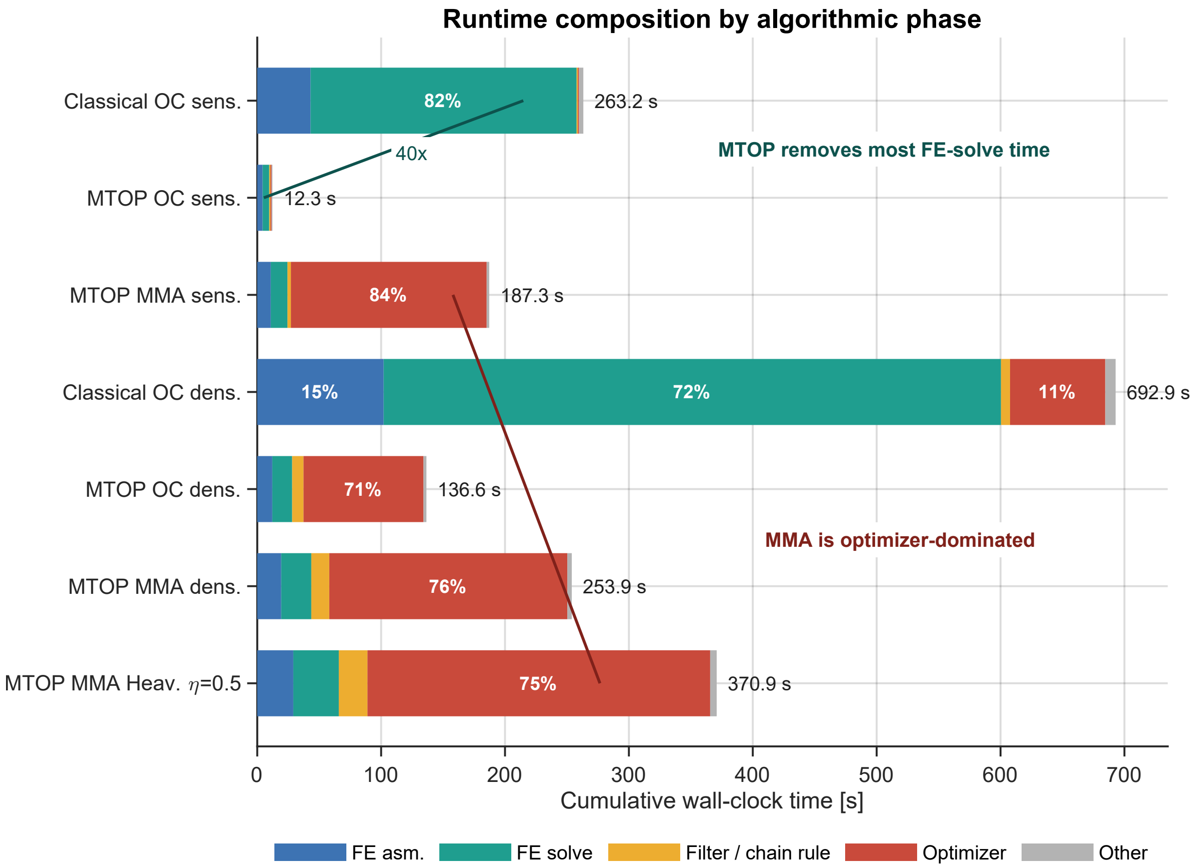


Figure 28: Per-phase wall-clock time, summed over all iterations. Each horizontal bar is one complete optimization run, decomposed into FE assembly, FE solve, filtering or chain-rule work, optimizer update and residual overhead. The chart highlights the two dominant timing effects: MTOP collapses the FE-solve cost for OC sensitivity filtering, while the MMA runs are dominated by the optimizer subproblem rather than by the equilibrium solve.

fig. 28 shows that, for sensitivity filtering, the FE solve itself is about 40 times faster in the MTOP run (214.77/5.35), while the total optimization is 21.4 times faster. The difference comes from the parts of the algorithm that do not shrink with the analysis mesh. The FE assembly speedup is lower because MTOP evaluates 25 stiffness contributions per analysis element, and the filter and OC update operate on the same 600×200 density field in both formulations. With density filtering this effect is stronger: the extra chain-rule convolution and the filtered-volume check add fixed per-density-cell work, reducing the overall MTOP speedup to 5.07 even though the equilibrium solve is still much cheaper.

8.4 Compliance evaluation and coarse-resolution comparison

The MTOP coarse-mesh compliance (219.069) is approximately 2.5 % below the fine-mesh compliance of the same design (224.696). This gap is not an optimization error; it comes from evaluating the stiffness with one midpoint contribution per density cell in eq. (7), instead of the analytical Q4 stiffness used in the classical fine-mesh model. The fine-mesh re-analysis is therefore the correct quantity for comparing the mechanical quality of the final designs. For the density-filtered runs the same offset is slightly smaller (2.3 %, from 235.363 to 240.984), because the filter pre-smooths the density field before the SIMP analysis.

Following the feedback discussion, a third reference case was added to separate the benefit of the coarse analysis mesh from the benefit of retaining a fine design description. This coarse-resolution baseline uses a classical 120×40 analysis mesh and the same 120×40 density mesh. Its filter radius is scaled to $r_{\min} = 4.8$ coarse elements, matching the physical radius of 24 density cells used in the 600×200 runs. The final coarse density field is then upsampled by a factor 5×5 and re-analyzed on the same 600×200 finite element mesh used for all final compliance comparisons.

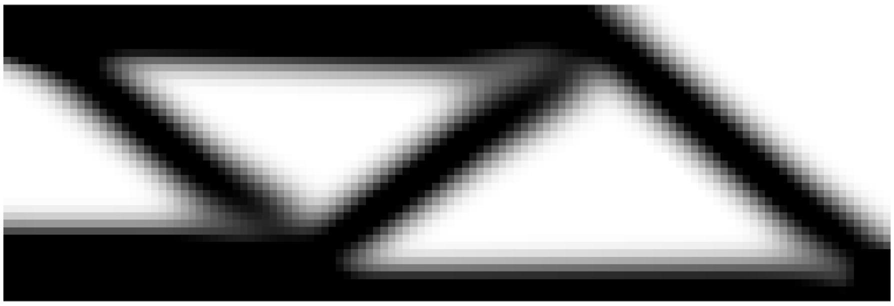


Figure 29: Upsampled coarse-resolution OC design optimized on a 120×40 analysis-density mesh with a physically equivalent filter radius $r_{\min} = 4.8$ coarse elements.

Table 5: Native and fine-mesh compliance comparison for the classical fine baseline, the MTOP sensitivity-filtered design and the additional coarse-resolution baseline. The fine-mesh compliance is the comparison value.

Model	Analysis mesh	Density mesh	Native C	Fine-mesh C
Classical fine	600×200	600×200	224.596	224.596
MTOP	120×40	600×200	219.069	224.696
Classical coarse-resolution	120×40	120×40	219.108	226.036

The coarse-resolution design has a fine-mesh compliance only 0.64 % above the classical fine baseline and 0.60 % above the MTOP design, despite using only 120×40 design variables. This shows that, for the present 2D half-MBB case and the relatively large filter radius used here, the main load path is already captured well by a much coarser model. The comparison is nevertheless not equivalent to MTOP: the coarse-resolution baseline has a higher grayness index, $\langle 4\rho(1-\rho)\rangle = 0.265$, so part of its low compliance comes from a smoother and less binary material distribution. MTOP keeps the fine 600×200 design description and nearly matches the full classical compliance while still solving the cheaper 120×40 equilibrium problem.

8.5 Density filtering and convergence behavior

For density filtering, both the classical and MTOP OC runs fail to satisfy the design-change tolerance $\varepsilon = 0.01$: the maximum absolute change oscillates between 0.04 and 0.07 while the compliance is already stable to well within 0.1 % of its final value. The cause is the incompatibility between the multiplicative OC update and density filtering on a fine mesh. The filter spreads each design-variable change over a neighbourhood of cells, but the OC update still rescales every variable individually, so high-frequency boundary changes can alternate from one iteration to the next. The compliance-stability criterion of Section 4.5 therefore gives a more meaningful stopping condition for these runs than the raw design-change criterion.

Density filtering also changes the MTOP comparison itself. The compliance gap between the classical and MTOP layouts remains very small: the fine-mesh compliance differs by only 0.04 %, and the design differences are mainly boundary-level changes. This happens because the density filter smooths the physical density before it enters the SIMP analysis, making the exact integration scheme inside each analysis element less influential. The price is computational: the density-filter chain rule and the filtered-volume check introduce additional cone convolutions on the full density mesh, so the MTOP speedup drops from 21.4 for sensitivity filtering to 5.07 for density filtering.

8.6 MMA and Heaviside projection

The MMA experiments of step 4 share the same MTOP analysis stack as the OC runs but use the method of moving asymptotes for the design update. The main observations are as follows.

First, MMA does not by itself fix the change-tolerance issue of density filtering. Both the OC and MMA density-filtered runs are terminated by the compliance-stability criterion rather than by $\max|\Delta x| < \varepsilon$, with similar residual oscillation. MMA produces a marginally better design on the fine mesh, but it does not damp the high-frequency boundary changes enough to satisfy the design-change tolerance.

Second, MMA does converge for sensitivity filtering, where OC also converged. The change drops below 0.01 after 250 iterations and the fine-mesh compliance is within 0.8 % of the MTOP OC result. The extra iterations are not surprising, because the sensitivity filter modifies the gradient heuristically rather than through an exact chain rule, which is less natural for MMA than for the monotonic OC update.

The clearest benefit of MMA appears when density filtering is combined with Heaviside projection. The projection creates a near-binary physical density field, and the continuation strategy in β lets MMA handle the increasingly sharp relationship between design variables and projected densities. For the symmetric threshold $\eta = 0.5$, the fine-mesh compliance drops from 240.077 for the MMA density-filtered design to 199.326, a reduction of about 17 %. The improved compliance comes from removing much of the gray material that is penalized inefficiently by SIMP.

The threshold η controls the bias of the projection. For $\eta < 0.5$, a lower filtered density is sufficient to project a cell to solid, so the projection tends to preserve solid regions; for $\eta > 0.5$, the projection is biased toward void. In this benchmark the effect on the final compliance is small: the three Heaviside fine-mesh compliances differ by less than 0.7 %, with 200.016 for $\eta = 0.3$, 199.326 for $\eta = 0.5$ and 200.633 for $\eta = 0.7$. The symmetric threshold gives the lowest value, but all three projected designs follow essentially the same load path. The main computational drawback is that the MMA subproblem becomes the dominant cost, as shown in fig. 28.

9 Conclusion

For the assigned two-dimensional MBB benchmark, MTOP is computationally efficient without sacrificing the optimized load path. With sensitivity filtering, the 120×40 MTOP analysis mesh and 5×5 density cells per analysis element reproduce the 600×200 classical design to within 0.045 % of fine-mesh compliance, while reducing the optimization time from 263.23 s to 12.28 s (21.4×). All final compliance comparisons are based on fine-mesh re-analysis, since the native MTOP compliance is slightly lower because of the coarse-analysis integration. The phase breakdown confirms that the speedup comes mainly from the equilibrium solve, which is about 40× faster in the MTOP run; the gain is partly absorbed by filtering, assembly and update operations that still act on the full density mesh.

The additional coarse-resolution baseline shows why this result should be interpreted carefully. A classical 120×40 analysis-density model, upsampled and re-analyzed on the fine mesh, reaches a compliance of 226.036, only 0.60 % above the MTOP design. For this 2D case with a large filter radius, a coarse model already captures the main load path well. The practical value of MTOP is therefore not only the compliance value, but the fact that it retains the fine 600×200 design description while solving the cheaper coarse equilibrium problem.

With density filtering, MTOP remains mechanically accurate but becomes less computationally advantageous. The classical and MTOP density-filtered layouts differ by only 0.04 % in fine-mesh compliance, but the speedup drops to 5.07 because the density-filter chain rule and filtered-volume check add work on the full density mesh. The density-filtered OC runs also do not satisfy the design-change tolerance; they are better interpreted as compliance-converged designs with a small residual high-frequency boundary oscillation.

Replacing OC by MMA does not remove this density-filter convergence issue by itself. MMA with sensitivity filtering gives a design close to the MTOP OC result, and MMA with density filtering still requires the compliance-stability stopping criterion. The strongest benefit of MMA appears when it is combined with Heaviside projection: the projected designs become nearly black-and-white, and the symmetric threshold $\eta = 0.5$ gives the best fine-mesh compliance, 199.326, about 17 % lower than the gray MMA density-filtered design. The three tested thresholds differ by less than 0.7 %, so their influence is small for this geometry, volume fraction and filter radius.

References

[1] E. Andreassen, A. Clausen, M. Schevenels, B. S. Lazarov, and O. Sigmund. Efficient topology optimization in MATLAB using 88 lines of code. *Structural and Multidisciplinary Optimization*, 43(1):1–16, 2011.

[2] T. Nguyen, G. H. Paulino, J. Song, and H. L. Chua. A computational paradigm for multiresolution topology optimization (mtop). *Structural and Multidisciplinary Optimization*, 41(4):525–539, 2010.

[3] O. Sigmund. A 99 line topology optimization code written in MATLAB. *Structural and Multidisciplinary Optimization*, 21(2):120–127, 2001.

[4] K. Svanberg. The method of moving asymptotes – a new method for structural optimization. *International Journal for Numerical Methods in Engineering*, 24(2):359–373, 1987.

[5] F. Wang, B. S. Lazarov, and O. Sigmund. On projection methods, convergence and robust formulations in topology optimization. *Structural and Multidisciplinary Optimization*, 43(6):767–784, 2011.

Code of conduct and transparency statement on the use of GenAI for KU Leuven-students (academic year 2025–2026)

Generative AI (GenAI) assistance tools can be used to generate various types of content, including text, images, code, video, music, or combinations thereof. Common examples of such tools include ChatGPT, Google Gemini, Microsoft Copilot, Midjourney, Claude.ai, Perplexity.ai, and DALL · E, among others.

This code of conduct is a tool that helps students to be transparent about the use of GenAI and fits within the university's principles on academic integrity.

Important guidelines and remarks

- **Sensitive or personal data:** Some GenAI tools protect your input and sensitive or personal data better than others. There is often no transparency on what the owners of the AI applications do with the data entered. Therefore, **do not enter sensitive or personal data in free GenAI tools.** More info about what sensitive and personal data are, is described in the three confidentiality levels of the KU Leuven data classification model (non-confidential, confidential, strictly confidential). For confidential data you should preferably use **M365 Copilot Chat with your KU Leuven account**. If you use another GenAI tool, you must be **absolutely certain** it does **not store or reuse the data you enter. Try to avoid entering these data. Do so only if strictly necessary, and only to the extent required. For personal data, work with anonymous or pseudonymized data.** Additionally, in case of strictly confidential data this should first be discussed with the teaching staff of the course or your thesis supervisor.
- **Copyrighted materials:** For **lawfully obtained copyrighted material** you should preferably use M**365 Copilot Chat with your KU Leuven account.** If you use another GenAI tool, you must be **absolutely certain** it does n**ot store or reuse the data you enter.**
- GenAI assistance may not be used for data or topics covered by a Non-Disclosure Agreement (NDA). Even in the absence of an NDA, certain information may still need to be treated as confidential—for example, due to regulatory requirements or the risk of significant harm to the university if disclosed. In case of doubt, check with your teaching staff or supervisor.
- If your master's thesis is under embargo, you should first discuss with your supervisor whether the use of GenAI is permitted.
- Before using a GenAI tool, always consider whether its use is responsible, including from a sustainability perspective (e.g. when using GenAI as a search engine, language assistant,...).
- **Take a scientific and critical attitude** when interacting with GenAI assistance and interpreting its output, that may not always be correct.
- As a student you are responsible for complying with Article 84 of the Regulations on Education and Examinations: your report or thesis should reflect your own knowledge, understanding and skills. Be aware that plagiarism rules also apply to (work that is the result of) the use of GenAI assistance tools.
  Exam Regulations Article 84: *"Every conduct individual students display with which they (partially) inhibit or attempt to inhibit a correct judgement of their own knowledge,*

understanding and/or skills or those of other students, is considered an irregularity which may result in a suitable penalty. A special type of irregularity is plagiarism, i.e. copying the work (ideas, texts, structures, designs, images, plans, codes , ...) of others or prior personal work in an exact or slightly modified way without adequately acknowledging the sources. Every possession of prohibited resources during an examination (see article 65) is considered an irregularity."

- In order to maintain academic integrity and avoid plagiarism, **more information about being transparent on the use of GenAI assistance and about correctly citing and referencing GenAI** can be found on this website for students (Dutch/English).
- **Additional reading: KU Leuven guidelines on responsible use of Generative AI tools, and other information** (Dutch/English)

A few final words

If you are uncertain whether or not you should declare your use of GenAI tools, we suggest that you discuss this with your instructor or supervisor. It is always safer to declare GenAI use, even when it is not strictly required. However, declaring GenAI use does not entail that its use is allowed; the right column in the table below provides more detailed instructions in this regard (code of conduct).

Moreover, advanced AI tools are evolving rapidly, and their capabilities have expanded significantly in a short period of time. As a result, we do not yet have all the answers about their responsible use. Finally, it is important to follow-up on the most recent evolutions in AI technologies, to have an open mind but also to be a bit cautious, to communicate with instructors, teaching assistants, supervisors and peers, to be as transparent as we can, and to learn together as we move along.

Student name: Lukas Campaert **Student number:** r0885501
Student name: Tiebert Lefebure **Student number:** r0887630

Course assignment information:

X This form is related to a **course assignment.**

Course name: Structural Optimization **Course number:** B-KUL-H0T88A

Please indicate with "X":

O I **did not use** any GenAI assistance tool.

X I **did use** GenAI Assistance. In this case **specify which ones** (e.g. ChatGPT, M365 Copilot, ...): ChatGPT

GenAI assistance used as/for:	**Name of the GenAI tool(s) used** If helpful, also describe in which way you were using GenAI related to what is specified as code of conduct.	**Code of conduct:** **For each of the categories below, always take into account the important guidelines and remarks mentioned above (e.g. copyrighted data, sensitive or personal data,...).**
As a language assistant for reviewing or improving texts I wrote myself	ChatGPT. Used to proofread paragraphs we wrote ourselves: catching typos, fixing grammar, and flagging awkward or Dutchified phrasings (literal Dutch-to-English constructions that read unnaturally in technical English). All edits were reviewed and rewritten by us before being kept.	This use is similar to using spelling and grammar check tools. In general, you do not have to refer to such kind of GenAI use in the text. **However, be careful:** • When using GenAI tools on texts you did not write yourself to improve the text, **you have to refer to the original source or author**, otherwise you are committing plagiarism and thus an irregularity.
As a paraphrasing tool	ChatGPT. Occasionally used to rephrase sentences and paragraphs we had written ourselves when our first draft felt clumsy or too long. Suggestions were treated as inspiration only: every paraphrased sentence was reread, edited, and rewritten in our own voice before being kept.	This use is allowed except when it is prohibited by the teacher or the program of study. You may paraphrase your own text or texts by an author other than yourself and take inspiration from what a GenAI tool or another tool suggests (unless it is not allowed). In general, you do not have to refer to such kind of GenAI use or other paraphrasing tools in the text. **However, be careful**: • If it entails text by an author other than yourself, **you are not allowed to include that paraphrased text without reference to the original source or author**. Without such reference, you would be committing plagiarism and thus an irregularity.

For translation aid to improve texts I wrote myself or to better understand text from others	No.	This use is allowed except when it is prohibited by the teacher or the program of study. It is similar to using translation tools (Google translate, DeepL, …). In general, you do not have to refer to such kind of GenAI use in the text. **However, be careful:** • **You are not allowed to include that translated text without reference to the original source or author.** Without such reference, you would be committing plagiarism and thus an irregularity. • Always check the translated text for correctness and meaning.
As a search engine to get information on a topic or to search for existing research on the topic	ChatGPT. Used as an initial aid to clarify how sensitivity verification is discussed in structural optimization; all information was checked against course material and cited sources before use.	This use is similar to e.g. a Google search or checking Wikipedia. If you write your own text based on this information, you do not have to refer to the use of GenAI in the text. **You only have to refer to the existing research and references you have checked and used** (without such references you would be committing plagiarism and thus committing an irregularity). **However, be careful:** • Be aware that the output of the GenAI tool cannot be guaranteed as a 100% reliable source of information. The output may not be entirely correct and/or be limited due to the databases it uses. Moreover, knowledge evolves and may change over time; therefore, the database of the GenAI tool may not be up to date. Therefore, **verify the information and do not just copy-paste it** as you should understand and critically process everything you are writing.

For literature search	No.	This use is comparable to e.g. a Google Scholar search. You do not have to refer to such kind of GenAI use; **you only have to refer to the literature references you have checked and used** (without such references you would be committing plagiarism and thus committing an irregularity). **However, be careful:** • Be aware that the search output is restricted to the database the GenAI tool is built on. After this initial search, look for scientific sources and conduct your own analysis of the source documents. **Interpret, analyse and process the information you obtained; verify it and do not just copy-paste it** as you should understand and critically process everything you are writing. • Be aware that some GenAI tools may **output no or wrong references.** As a student you are responsible for further checking and verifying the absence or correctness of references; do not just copy-paste it.
For generating programming code	ChatGPT was used only to support post-processing and logging scripts for the simulations; the simulation workflow and results were checked and validated by us.	Use of GenAI for coding is allowed except when it is prohibited by the teacher or the program of study. If used for coding, correctly mention the use of GenAI assistance in accordance with the instructions on the page on being transparant about the use of GenAI.
For generating new (research) ideas	No.	This use of GenAI is allowed except when it is prohibited by the teacher or the program of study. Correctly mention the use of GenAI assistance in accordance with the instructions on the page on being transparent about the use of GenAI. **Be careful:** • Further verify in this case whether the idea is novel or not. It is likely that it is related to existing work. **If so, that existing work should be correctly referenced in the text** (without such reference you would be committing plagiarism and thus committing an irregularity).

For generating synthetic data	No.	Use of GenAI for generating synthetic data is allowed, **provided that it is methodologically and ethically justifiable**, except when it is prohibited by the teacher or the program of study. Always correctly mention the use of GenAI assistance in accordance with the instructions on the page on being transparent about the use of GenAI. **Be careful:** • Always **carefully evaluate** the generated synthetic data for quality and possible bias since the output is highly dependent on the quality of the data on which the models are trained.
For generating blocks of text (other than the allowed use without referencing mentioned above)	No.	According to Article 84 of the Regulations on Education and Examinations your text should allow to correctly and properly assess your own knowledge, understanding and skills. **Therefore, inserting blocks of text without quotes and a reference to GenAI assistance in your work is not allowed.** **Be careful:** • If it is really needed to insert a block of text from a GenAI tool, for instance because of the nature of your assignment, mention it as a citation by using quotes and correctly mention the use of GenAI assistance in accordance with the instructions page on being transparent about the use of GenAI. • However, in general, such GenAI use should be kept to an **absolute minimum**; you should always check the original sources.

<table>
<tr><td>For generating visuals, video or audio</td><td>ChatGPT. Used to help generate supporting figures for theory, such as the MMA graph, and to visualize data obtained from our own simulations, such as the time-reduction comparison.</td><td>Use of GenAI for generating visuals, audio or video is allowed except when it is prohibited by the teacher or the program of study.

Be careful:
If used, refer to GenAI for visuals according to the usual referencing style, following the instructions on the page on referencing GenAI.
If you work with existing visuals, audio or video, that existing work should be correctly referenced in the text (without such reference you would be committing plagiarism and thus an irregularity).
Explain the usage in the methods section (if there is one) and optionally attach (or link to) the prompts with the full output (or history).</td></tr>
<tr><td>Other use (specify here; this may also include a combination of types of use mentioned above):</td><td>No.</td><td>To make sure other use of GenAI is allowed within the course or thesis, and if so, the conditions that may apply, contact the teaching staff of the course or the supervisor of the thesis beforehand and explain the intended GenAI use. Also inform the programme director.

Motivate how you would comply with Article 84 of the exam regulations. Explain the use and the added value of the AI tool you consider to use and how it is in accordance with the assignment or thesis and the KU Leuven guidelines on responsible use of Generative AI tools.

Depending on the kind of GenAI use, it may be needed to properly reference it in the text, in accordance with the instructions page on being transparent about the use of GenAI.</td></tr>
</table>